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Integrating generative AI into STEM education: enhancing conceptual understanding, addressing misconceptions, and assessing student acceptance

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Abstract

Advancements in artificial intelligence (AI), particularly generative AI models such as ChatGPT, offer transformative opportunities to enhance educational practices in STEM disciplines. Thermodynamics, a fundamental subject in engineering education, presents significant challenges due to its abstract nature and common misconceptions. This study investigates the effectiveness of integrating ChatGPT as a supplemental pedagogical tool, guided by a constructivist inquiry-based approach using the Constructivist Inquiry-Based Learning Prompting (CILP) framework, to enhance conceptual understanding and address misconceptions in an introductory thermodynamics course for first-year Moroccan engineering students. A quasi-experimental design was used, with 120 students equally divided into control and experimental groups. The control group received traditional instruction, whereas the experimental group received ChatGPT-assisted instruction. Conceptual understanding was measured using pre- and post-tests, while student perceptions and acceptance were collected via weekly surveys. Results showed that the experimental group significantly outperformed the control group, exhibiting greater improvements in conceptual understanding and a reduction in qualitative misconceptions, particularly related to entropy and internal energy. However, some quantitative misconceptions persisted, underscoring ChatGPT's limitations in advanced reasoning tasks, problem-solving, and numerical calculations. Students reported high satisfaction with ChatGPT's usability and instructional support. Moreover, targeted use of ChatGPT, rather than frequent reliance, correlated with optimal learning outcomes. These findings underscore ChatGPT's potential to enhance STEM education within inquiry-based, constructivist learning environments and provide evidence for the effective integration of generative AI tools to improve learning outcomes, particularly in resource-constrained settings.

Keywords Generative AI, STEM education, Misconceptions, Conceptual understanding, Inquiry-based learning

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Introduction

Recent advancements in artificial intelligence (AI) have been driven by significant developments in three critical domains: the evolution of neural network architectures, the refinement of machine learning algorithms, and the proliferation of deep learning methodologies (Alzubaidi et al., 2021; Mienye & Swart, 2024). These technological innovations have culminated in the emergence of large language models (LLMs), which have substantially enhanced AI's capacity to process, generate, and interact with natural language in more sophisticated ways. Generative AI models, such as ChatGPT, Gemini, and Claude, demonstrate remarkable proficiency across diverse tasks, including conversational engagement, content creation, complex problem-solving, and data analysis (Kasneci et al., 2023; Akhtar, 2024). Consequently, these developments are reshaping education by transforming pedagogical practices, supporting personalized learning approaches, and redefining the broader educational experience.

Building on these innovations, AI-powered tools offer unprecedented opportunities to enrich teaching and learning processes. Such tools facilitate personalized learning experiences, provide immediate feedback, and deliver adaptive instructional support tailored to the diverse needs of learners (Chan & Hu, 2023; Eager & Brunton, 2023). For instance, generative AI models like ChatGPT simulate human-like tutoring, clarify complex concepts, and engage students in interactive dialogues that promote deeper understanding (Gregorcic & Pendrill, 2023; Getachew, 2024). Unlike pre-scripted educational resources, such as PhET simulations or virtual reality (VR) labs, which are valuable yet limited by static content (Banda & Nzabahimana, 2023; Gagnon et al., 2024; Santilli et al., 2025), generative AI offers dynamic and adaptive scaffolding through real-time, dialogic interactions. By addressing misconceptions in real time, generative AI bridges conceptual gaps and resolves challenges in STEM education.

The transformative potential of AI in education is substantiated by recent research. Studies show that AI models based on Transformer architectures can diagnose student misconceptions in physics with accuracy comparable to human experts (Demirezen et al., 2023). Similarly, advanced systems such as the Enhanced Ensemble Model have proven effective in addressing conceptual misunderstandings, presenting alternatives to traditional pedagogical methods (Kökver et al., 2024). These findings highlight AI's ability to enhance educational outcomes, particularly in STEM disciplines where conceptual clarity is essential.

However, students continue to encounter persistent cognitive challenges in STEM subjects, including misconceptions and foundational knowledge gaps. These

challenges are especially pronounced in Morocco, where significant efforts are underway to assess and reform the national education system. National and international assessments, such as the National Programme for Student Learning Assessment (PNIA), the Programme for International Student Assessment (PISA), and the Trends in International Mathematics and Science Study (TIMSS), have revealed substantial gaps in mathematics and science knowledge, particularly among first-year students (INE, 2021; OECD, 2023a; Mullis et al., 2020). Addressing these gaps is critical to fostering academic progress and equipping students with the advanced problem-solving and critical thinking skills increasingly demanded by STEM fields.

Thermodynamics, a cornerstone of engineering curricula worldwide, exemplifies these challenges. This foundational discipline is vital for understanding energy systems, materials science, and mechanical processes, yet it remains one of the most difficult courses due to the abstract nature of its concepts and the complexity of its mathematical frameworks. Research has consistently highlighted widespread student misconceptions, such as interpreting entropy merely as a measure of disorder or misapplying the second law of thermodynamics (Brundage & Singh, 2023a; Brundage et al., 2024). These enduring misconceptions underline the limitations of traditional teaching methods and underscore the need for innovative, interactive strategies to enhance engagement and understanding. In Morocco, these challenges are compounded by a reliance on traditional lecture-based teaching methods, which often fail to address individual learning needs. Generative AI offers a promising solution by providing real-time feedback, adaptive responses, and personalized dialogue, thus addressing misconceptions and fostering deeper conceptual understanding (Sands, 2024).

The global emphasis on ethically deploying AI in education, as advocated by organizations such as UNESCO and the OECD, aligns with the potential of generative AI tools to address individual learning needs (Emma & Arianna, 2023; Miao & Holmes, 2023; OECD, 2023b). In Morocco, where empirical research on generative AI models remains scarce, this alignment is particularly relevant. Morocco's educational system, characterized by rigorous engineering preparatory programs such as the Preparatory Classes for Higher Education Institutions (CPGE), emphasizes engineering and STEM disciplines (Martin Rose, 2017). This context presents both challenges and opportunities for AI integration. Given the persistent difficulties students face in mastering thermodynamics, there is an urgent need to investigate how AI tools can be effectively leveraged to improve learning outcomes in this domain.

Study purpose and research questions

This study investigates the efficacy of integrating ChatGPT, a generative AI chatbot, as a supplementary pedagogical tool in an introductory thermodynamics course for first-year Moroccan engineering students. The primary objectives are to enhance students' conceptual understanding of thermodynamic principles, address common misconceptions, and foster a more interactive and engaging learning environment. Additionally, the study evaluates students' perceptions and acceptance of ChatGPT in terms of usability, instructional support, and overall effectiveness as a classroom tool.

While AI tools like ChatGPT are increasingly being integrated into education, limited research explores their application in engineering curricula or their potential to address subject-specific challenges, such as misconceptions in thermodynamics. This study aims to address these gaps by contributing to global discourse on the integration of AI in STEM education and offering a framework for applying generative AI tools in classroom settings. By focusing on Moroccan engineering curricula, the study also provides insights into region-specific challenges and opportunities for leveraging AI in education. To systematically explore these objectives, the study is guided by the following research questions (RQs):

RQ1 How does integrating ChatGPT as a supplementary pedagogical tool enhance students' conceptual understanding and address their misconceptions about thermodynamic topics?

RQ2 How do students perceive and accept ChatGPT's role in facilitating their understanding of thermodynamics, particularly in terms of usability, instructional support, and overall effectiveness?

RQ3 What is the impact of integrating ChatGPT into classroom activities over a semester on student engagement and overall learning outcomes in thermodynamics? This study ultimately seeks to determine whether ChatGPT and similar generative AI tools can be effectively integrated into existing classroom environments to improve learning outcomes, deepen understanding of complex scientific concepts, and foster widespread acceptance among students.

Literature review

Generative AI in STEM pedagogy

Generative AI, particularly LLMs such as ChatGPT, has significantly transformed education by facilitating personalized and adaptive learning experiences. These tools have leveraged context-sensitive responses and dynamic feedback to facilitate engagement with complex educational content and support diverse activities such as

writing, brainstorming, research, and critical analysis (Chan & Hu, 2023; Eager & Brunton, 2023; O'Dea, 2024). For example, ChatGPT's ability to simulate Socratic questioning and provide real-time feedback has been shown to foster inquiry-based learning and deepen critical thinking in physics and chemistry education (Gregorcic *et al.*, 2024).

The effectiveness of generative AI tools is largely dependent on the quality of user-generated prompts, as precise input directly influences the relevance and accuracy of output. Despite their potential, generative AI systems often exhibit limited nuanced contextual understanding, which poses challenges for deeper inquiry. However, their application in philosophical dialogues has demonstrated the capacity to enhance inquiry-based discussions and promote meaningful classroom engagement (Getachew, 2024; Gregorcic & Pendrill, 2023). In the context of physics education, tools like ChatGPT have proven effective in addressing common misconceptions and offering immediate, tailored feedback, particularly in intricate topics such as thermodynamics (Kieser *et al.*, 2023; Kieser & Wulff, 2024). These findings position generative AI as a versatile tool for addressing persistent learning challenges across STEM disciplines.

Benefits and challenges of AI integration in STEM education

The integration of generative AI into STEM education has received considerable attention due to its transformative potential for teaching and learning. Research indicates that generative AI enhances student engagement and learning outcomes in fields such as mathematics, physics, and engineering (Chan & Hu, 2023; Polverini & Gregorcic, 2024). For instance, tools like ChatGPT have delivered real-time feedback to rectify misconceptions, thereby fostering interactive learning experiences and facilitating deeper conceptual understanding (Labadze *et al.*, 2023; Moro-Visconti, 2024). Additionally, these systems can customize learning pathways to meet individual needs, making them particularly valuable in fields characterized by diverse skill sets, such as mathematics and engineering (Polverini & Gregorcic, 2024).

Despite these advantages, challenges such as inherent AI biases, data privacy concerns, and infrastructural limitations have posed barriers to widespread adoption. Ensuring trust and security is critical in educational environments, where the successful integration of AI depends on its acceptance by both students and educators. Key factors influencing trust and adoption include perceived ease of use, utility, and intelligence (Shahzad *et al.*, 2024; Yang & Appleget, 2024). Furthermore, cultural contexts significantly shape adoption processes through performance expectancy, effort expectancy, and social

influence (Budhathoki et al., 2024; Farhi et al., 2023; Hsu & Ching, 2023).

Over-reliance on AI tools could potentially undermine essential skills such as critical thinking and problem-solving, which are foundational to STEM education and real-world innovation (Chen et al., 2023). Therefore, achieving a balance between AI-enhanced learning and traditional teaching methodologies is vital to cultivating independent thinking and problem-solving capabilities (Popenici & Kerr, 2017; Sandu et al., 2024). Generative AI offers the potential for creating personalized, engaging, and efficient learning experiences, but addressing its associated challenges necessitates curriculum redesign and targeted professional development to align AI use with educational objectives. Strategic implementation of AI-driven pedagogical frameworks could bridge longstanding learning gaps in STEM disciplines.

Innovative pedagogical approaches in STEM education

Thermodynamics, often regarded as an abstract and conceptually challenging subject, has benefited from a variety of pedagogical strategies aimed at improving student learning outcomes and engagement (Aidoo, 2024). Constructivist approaches, such as the Energy Chain Model, prioritize active learning and the integration of qualitative and quantitative reasoning to address persistent misconceptions in thermodynamics (Meli et al., 2022).

Gamification strategies have enhanced collaboration and engagement but may prioritize surface-level interaction over deeper conceptual understanding. For example, gamified modules in thermodynamics have been shown to increase student participation but often require supplementary instruction to fully address underlying principles (Suárez-López et al., 2023).

Simulation-based learning environments, such as PhET simulations, have supported inquiry-based learning and critical thinking by allowing students to visualize thermodynamic phenomena. These tools enable variable manipulation, aiding in the exploration of concepts such as gas behavior, pressure-volume relationships, and particle interactions (Almasri, 2022; Banda & Nzabahimana, 2023; Dorji et al., 2024). Computational tools, such as MATLAB Grader, further support active problem-solving in advanced topics, including numerical modeling and equations of state, while providing real-time feedback (Camacho-Lie et al., 2024).

Emerging technologies like Virtual Reality (VR) and Augmented Reality (AR) have introduced immersive dimensions to thermodynamics education. For instance, ThermoVR incorporates embedded assessments to simulate piston-cylinder systems, and Physics Holo.lab overlays real-time thermal data on experimental setups, facilitating visualization and immediate feedback (Strzys

et al., 2018; Pfortenhauer & Gagnon, 2021; Gagnon et al., 2024).

These innovative teaching methods highlight the potential to make thermodynamics more accessible and intellectually enriching. Generative AI complements these strategies by addressing conceptual gaps, fostering interactive learning, and enhancing student-centered educational experiences.

Leveraging AI to enhance STEM education

Generative AI presents numerous opportunities to bridge the gap between theoretical knowledge and practical application in STEM education (Aslam et al., 2023; Dare et al., 2021). In physics, for example, AI tools effectively address common misconceptions and deliver real-time, personalized feedback to enhance students' conceptual understanding (Elmas et al., 2024; Kotsis, 2024).

To fully realize the potential of AI in STEM education, substantial investments are required in curriculum redesign, educator training, and the development of effective integration strategies (Bernabei et al., 2023; Park et al., 2023). Aligning AI tools with traditional educational goals necessitates targeted professional development to prepare educators for AI-enhanced teaching environments (Dare et al., 2021; Gregorcic & Pendrill, 2023). Developing AI-supported curricula and teacher training programs is essential to bridge the gap between AI capabilities and classroom objectives (Shernoff et al., 2017; Gregorcic et al., 2024).

Bridging gaps in AI-Driven STEM education

While generative AI offers substantial opportunities for advancing STEM education, significant challenges remain, particularly in under-resourced settings. Infrastructural limitations, such as restricted access to technology and inadequate teacher training, have hindered the widespread adoption of AI tools. Further research is needed to explore AI applications across diverse global contexts, especially in low- and middle-income countries such as Morocco.

Morocco's CPGE program provides a unique case for studying AI's impact on STEM education. Emphasizing mathematics, physics, and engineering (CNIPE) 2012, the CPGE curriculum has offered valuable insights into AI adoption in resource-constrained environments (Assad & Ait Si Mhamed, 2020; Martin Rose, 2017). This study examines the impact of AI tools like ChatGPT on learning outcomes in thermodynamics within Morocco's Mathematics, Physics, and Engineering Sciences (MPSI) and Physics, Chemistry, and Engineering Sciences (PCSI) tracks. Additionally, it investigates factors influencing AI acceptance among Moroccan engineering students, offering critical insights into the opportunities and challenges of implementing AI in under-resourced settings.

In conclusion, generative AI holds transformative potential for STEM education, offering personalized learning experiences, real-time feedback, and innovative teaching tools. However, addressing challenges such as biases, data privacy concerns, and infrastructural limitations is essential to ensure equitable access and effective implementation. Integrating evidence-informed traditional approaches with innovative AI-driven strategies is crucial to fostering critical thinking and deep conceptual understanding in STEM disciplines.

Materials and methods

Research design

This study employed a quasi-experimental design with continuous assessment to evaluate the impact of ChatGPT, a generative AI chatbot, as a supplementary pedagogical tool in an introductory thermodynamics course for Moroccan engineering students. Participants were divided into two groups: a control group that adhered to traditional instructional methods and an experimental group that incorporated ChatGPT into their learning activities. The research aimed to measure improvements in students' understanding of key thermodynamic concepts and assess ChatGPT's effectiveness in addressing common misconceptions using a pretest-posttest design. Additionally, a continuous assessment approach was integrated through the Technology Acceptance and Learning Enhancement Survey (TALES), which collected longitudinal data on student engagement, perceptions of ChatGPT's usefulness, and its impact on learning outcomes over the semester. This comprehensive research design sought to provide detailed insights into the potential of generative AI tools, specifically ChatGPT, to enhance educational practices while maintaining rigorous ethical standards within the Moroccan engineering curriculum.

Participants

The study was conducted during the autumn semester of 2023, involving 120 first-year engineering students from the preparatory cycle of the engineering program at the National School of Applied Sciences, Ibn Tofail University, in Kenitra, Morocco. These students were selected through convenience sampling due to their accessibility and relevance to the study's objectives. The cohort consisted of 54 male and 66 female students, aged between 19 and 21 years. All participants were enrolled in an introductory thermodynamics course, a critical component of their engineering curriculum. Participation was entirely voluntary, and the study adhered to strict ethical standards, respecting participant autonomy, ensuring data confidentiality, and complying with established ethical guidelines.

Research instruments

Survey of thermodynamic processes and first and second laws

The primary instrument for this study was the Survey of Thermodynamic Processes and First and Second Laws (STPFSL), in its abbreviated form, STPFSL-Short. This multiple-choice survey rigorously assesses students' comprehension of fundamental thermodynamic principles and assists instructors in evaluating the effectiveness of innovative pedagogies in thermodynamics. The STPFSL-Short, a validated and reliable tool developed by Brown and Singh (2021), is a subset of the longer version, STPFSL-Long, which was further refined by Brundage and Singh (2023b). The STPFSL-Short consists of 33 items covering core thermodynamic principles. Following the International Testing Commission (ITC) guidelines (Hambleton, 2001), the survey was translated into French and verified for accuracy by subject matter experts. A pilot study was conducted to refine the survey. The French version of the STPFSL-Short is provided in Appendix C.

Technology acceptance and learning enhancement survey

To explore the dynamics of generative AI adoption and its impact on Moroccan engineering students' learning, the research team developed the TALES by drawing on theoretical constructs from the Technology Acceptance Model (TAM) family (Fred & Davis, 1989; Venkatesh & Davis, 2000; Venkatesh & Bala, 2008) and the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003), to align with the study's objectives. As a secondary measurement instrument, TALES evaluates students' perceptions and acceptance of generative AI tools, specifically ChatGPT, as a pedagogical resource within the learning environment. The survey assesses key dimensions of AI acceptance, including perceived usefulness, perceived ease of use, behavioral intention, and usage behavior, with a particular focus on thermodynamics to evaluate the behavioral impact of the intervention.

The 16-item survey combines five-point Likert scales and multiple-choice questions, focusing on the frequency and context of ChatGPT usage, its role in addressing misconceptions and its effects on learning processes, including conceptual development and confidence enhancement throughout the semester (see Appendix A). By incorporating TALES as a weekly survey during the study's procedural phases, the research team maintained rigorous standards for continuous assessment and monitored trends in students' technology acceptance. This approach ensured a systematic data collection process, essential for evaluating how generative AI tools influence students' cognitive engagement, conceptual understanding, and technology acceptance in STEM education.

Constructivist Inquiry-Based learning prompting framework

The CILP framework is presented as a novel pedagogical model that integrates core principles of constructivist learning theory (Lebow, 1993; B.G.W, 1996). It is operationalized through inquiry-based learning and prompt engineering to enhance the integration of generative AI tools in STEM education. Grounded in the premise that learners construct knowledge most effectively through active engagement and iterative reflection, the CILP framework extends beyond procedural guidance by embedding theoretical principles into prompt design. Unlike traditional prompt engineering, the CILP framework emphasizes self-regulated learning, metacognitive reflection, and critical assessment, ensuring that learners refine AI queries while simultaneously enhancing their cognitive processes.

Moreover, the CILP framework is flexible and seamlessly integrates into existing STEM curricula, reinforcing inquiry-driven exploration without imposing rigid templates or instructions. By iteratively refining prompts, evaluating generative AI outputs, and engaging in a continuous cycle of questioning, assessing, and reflecting, it transforms generative AI from a passive answer provider into an active learning partner. This shift empowers learners to interact with AI outputs in ways that deepen conceptual understanding, cultivate critical thinking, and strengthen problem-solving skills. Key objectives in STEM disciplines such as physics, chemistry, and engineering.

In practice, instructors and students adopting the CILP framework begin by formulating open-ended, inquiry-driven prompts aligned with specific learning objectives. Students then engage with AI-generated responses, critiquing and refining both the prompts and their cognitive processes based on feedback. This reflective process fosters metacognitive awareness, enabling learners to identify misconceptions, adjust their approaches, and build stronger conceptual connections. The CILP framework positions generative AI as a dynamic tool for advancing inquiry, collaboration, and conceptual growth in STEM teaching and learning.

Procedures

Preparation phase

This study focused on first-year engineering students at the National School of Applied Sciences during the first semester of the 2023–2024 academic year. The participants were enrolled in an introductory thermodynamics course and had no prior formal instruction in thermodynamics, ensuring a common baseline for assessing their conceptual understanding. To maintain instructional consistency, the students were divided into experimental and control groups with equivalent academic profiles,

including similar entrance examination scores and prior academic performance. Both groups were supervised by the same instructor.

Before the intervention, all students completed the STPFSL-Short pretest (40–45 min) under standardized classroom conditions to assess their initial conceptual understanding, identify common misconceptions, and establish a baseline for comparison. The experimental group participated in a structured, three-hour training session on the CILP Framework (see Appendix A). This session operationalized the CILP Framework by aligning activities with its core principles, such as inquiry-based learning, iterative refinement of prompts, and metacognitive reflection. Students engaged in case-based scenarios that required them to design, craft, and refine prompts aligned with the thermodynamics coursework. They practiced evaluating ChatGPT's responses for accuracy and relevance, refining their approaches based on feedback, and collaboratively analyzing the strengths and limitations of generative AI tools in solving thermodynamic problems.

Implementation phase

The intervention was implemented during tutorial sessions, which were extended by 30 min each week, resulting in two-hour tutorials totaling 32 h over the semester. In the experimental group, these extended sessions emphasized applying the CILP Framework principles. Students actively crafted, refined, and evaluated prompts aligned with specific course objectives. Tutorials were conducted in a technology-enhanced classroom (computer lab) equipped with reliable internet access and computers. Students used the free version of ChatGPT (versions 3.5 and later), accessed through the OpenAI platform, as a supplemental pedagogical resource. The additional 30 min of each session were dedicated to activities such as designing and refining prompts, analyzing AI-generated responses for accuracy and depth, and participating in instructor-led discussions. These discussions addressed conceptual challenges, clarified misconceptions, and explored thermodynamic principles in real-world scenarios. The instructor provided real-time feedback on the quality of prompts and problem-solving approaches, while collaborative discussions reinforced both theoretical understanding and practical applications of thermodynamic principles.

In parallel, the control group followed the same tutorial schedule, meeting for two hours per week in a traditional physics classroom without access to technological tools. Their tutorials employed conventional methods, including instructor-led problem-solving, walkthroughs of worked examples, and guided practice, such as using instructor handouts and guided exercises. This approach ensured consistency in instructional intensity and time

allocation, isolating the use of generative AI as the primary variable. Both groups participated equally in lectures and laboratory sessions as part of the thermodynamics curriculum, ensuring equal exposure to all instructional components.

Monitoring and data collection phase

Throughout the intervention (8 week), students in the experimental group documented their experiences, challenges, and strategies for effectively leveraging ChatGPT. Weekly reflections, collected via the TALES survey using Microsoft Forms, captured evolving perceptions of ChatGPT across dimensions such as perceived usefulness, ease of use, and impact on learning processes. These reflections were analyzed to identify trends in AI acceptance and explore their relationship with students' conceptual growth over the semester. The instructor periodically reviewed these reflections to adjust instructional strategies, ensuring alignment with the CILP Framework principles and addressing emergent challenges. Simultaneously, the activities of the control group were monitored to ensure parity in instructional time and intensity, allowing any observed differences in learning outcomes to be attributed solely to the use of generative AI and the CILP Framework.

Post-Implementation phase

At the end of the instructional period, all students completed the STPFSL post-test under standardized conditions to evaluate their conceptual understanding and track changes in misconceptions. The results provided quantitative measures of learning gains and conceptual improvements across both groups. Additionally, the experimental group completed a final feedback survey, incorporating items from the TALES survey, to assess their learning experiences with ChatGPT and its influence on learning outcomes. This comprehensive evaluation linked students' subjective experiences with measurable outcomes, providing a robust assessment of the intervention's effectiveness in enhancing conceptual understanding, critical thinking, and problem-solving skills.

Data analysis

Following data collection from both the STPFSL and TALES, statistical analyses were conducted using R (version 4.4.1; R Core Team, 2024) and Microsoft Excel for data manipulation, visualization, and interpretation. Initially, the data were screened for missing values and outliers, which were subsequently addressed. Next, item difficulty and discrimination indices for the STPFSL were calculated using the psych package (Revelle, 2007), following guidelines based on Classical Test Theory (CTT; Crocker & Algina, 1986).

Principal Component Analysis (PCA) was conducted using the psych package to reduce the dimensionality of the data and to identify underlying factors within the STPFSL. Before factor extraction, Horn's parallel analysis was performed using the paran package (Alexis Dinno, 2007), to determine the optimal number of components to retain, ensuring that only components capturing substantive variance were selected. Subsequently, Promax rotation was applied to account for potential correlations among factors. The suitability of the dataset for PCA was confirmed by conducting the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's Test of Sphericity. Additionally, assumptions of linearity, sampling adequacy, and appropriate inter-item correlations were systematically verified.

Cluster analyses were conducted in two distinct phases to gain deeper insights into student performance and misconceptions regarding topics in thermodynamics, implemented using the cluster package (Maechler et al., 1999). First, a conceptual-understanding clustering involved applying the K-means clustering algorithm to students' pretest and posttest STPFSL scores to identify distinct patterns and trends in conceptual understanding. This process yielded three categories of understanding levels: low, moderate, and high. Second, a misconception segmentation involved a separate K-means analysis, examining the frequencies of incorrect responses on specific STPFSL items as indicators of misconception prevalence. This analysis identified three levels of misconceptions: high, moderate, and low. The optimal number of clusters was determined using the Elbow method, and Silhouette Analysis was employed to validate the quality and distinctiveness of cluster separations (Kaufman & Rousseeuw, 2005). This dual-clustering approach effectively categorized students based on their overall conceptual understanding and persistent misconceptions, providing insights into student subgroups, their specific learning needs, and the effectiveness of instructional interventions within the CILP framework.

For both cluster analyses, the distributions and shifts of clusters between the control and experimental groups were compared and examined. Chi-square tests were used to assess shifts in the frequencies of misconceptions. Additionally, descriptive statistics and correlation analyses were performed to examine students' perceptions measured by the TALES. Finally, a Generalized Linear Model (GLM), implemented via the stats and glm2 packages (Marschner, 2011), was employed to investigate how these perceptions influenced learning outcomes. This multifaceted analytical approach allowed for a comprehensive evaluation of the instructional intervention, identifying key factors contributing to learning gains in

thermodynamics, and highlighting students' perceptions of and acceptance of generative AI-enhanced instruction.

Reliability and validity analysis

To further assess the quality of the STPFSL items and their effectiveness in distinguishing between high- and low-performing students, item difficulty and discrimination indices were analyzed using a CTT approach. For the TALES, Cronbach's alpha was calculated for each dimension of the survey: Perceived Usefulness, Perceived Ease of Use, Behavioral Intention, and Usage Behavior. The resulting subscale reliability coefficients ranged from 0.581 to 0.701. An overall Cronbach's alpha coefficient of 0.693 indicated moderate internal consistency.

For STPFSL, reliability was deemed acceptable, with values ranging from 0.618 to 0.690. The Processes dimension exhibited higher reliability in the experimental group (Pretest EG: 0.784, Posttest EG: 0.823) than the control group, reflecting improved internal consistency due to the intervention. Other dimensions, such as Quantities and Relations, achieved high reliability (e.g., Posttest EG: 0.894). Lower reliability in some areas was attributed to the conceptual complexity of specific items within those subscales (see Table 1).

Following the confirmation of the reliability of the STPFSL and TALES instruments, their validity was assessed through content, face, and construct validity evaluations. The validity of TALES was established through content and face validity assessments (Holden, 2010). To evaluate content validity, three subject matter experts in physics education, educational technology, and AI from the National School of Applied Sciences reviewed the survey items. Their evaluation ensured alignment with theoretical constructs from TAM family and the UTAUT. The experts assessed the items' relevance, clarity, and coverage in relation to the study's objectives. Based on their feedback, minor revisions were implemented, including refining item wording for clarity and adding a question regarding students' trust in the

accuracy of generative AI-generated thermodynamics explanations.

Face validity was assessed through a pilot test involving 20 engineering students excluded from the main study. These participants evaluated the survey for clarity, relevance, and ease of understanding. Key feedback included suggestions to simplify technical language in some items and rephrase ambiguities. Insights from this pilot test were instrumental in refining the TALES survey.

The construct validity of the STPFSL was examined using PCA with Promax rotation, applied to pretest and posttest data from the control and experimental groups. PCA loadings identified distinct components aligned with key conceptual dimensions of thermodynamics (see Appendix B, Table B1-B4), providing evidence of construct validity. In the control group, three components explained 70.6% of the total variance, while the experimental group exhibited a single dominant component accounting for 71.4%. This single dominant component suggests stronger construct validity in the experimental group, potentially due to the intervention's influence. The Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy was 0.71 for the control group and 0.81 for the experimental group, indicating good to excellent sampling adequacy. Values of the KMO measure above 0.7 indicate that the dataset is suitable for PCA, while values above 0.8 represent excellent sampling adequacy. Bartlett's test of sphericity was significant for both groups (control group: $\chi^2(231) = 3695.47$, $p < 0.001$; experimental group: $\chi^2(231) = 5036.57$, $p < 0.001$), confirming the factorability of the correlation matrix.

To further establish validity, item difficulty and discrimination indices were analyzed using the CTT approach (DeVon et al., 2007). This analysis provided additional evidence of the STPFSL's criterion-related validity and indirectly supported its construct validity. The experimental group demonstrated a significant improvement in item difficulty (from 0.42 to 0.65), reflecting enhanced understanding due to the intervention. In comparison, the control group showed a smaller increase (from 0.35 to 0.55). Discrimination indices remained consistent, averaging 0.40 for the experimental group and 0.33 for the control group. These values indicate that the STPFSL items effectively differentiate between high and low performers, with better discrimination observed in the experimental group (see Appendix B, Figures B1-B4).

Table 1 Internal consistency metrics (Cronbach's Alpha)

Dimension	Pretest (CG)	Posttest (CG)	Pretest (EG)	Posttest (EG)
Processes	0.717	0.757	0.784	0.823
Systems	0.761	0.762	0.706	0.745
Quantities and Relations	0.871	0.897	0.877	0.894
Representation	0.363	0.575	0.329	0.547
First Law	0.494	0.517	0.474	0.571
Second Law	0.589	0.547	0.451	0.497
Overall measure	0.618	0.659	0.660	0.690

Note. CG=Control Group, EG=Experimental Group

Results

For RQ1

To address Research Question 1 (RQ1), we utilized a pretest–posttest design with STPFSL assessments to evaluate students' thermodynamics knowledge before and after the intervention. This design also allowed us to compare the learning progress between the control and

Table 2 Conceptual learning gains comparison across control and experimental groups

Measure	CG Mean (SD)	CG Range	EG Mean (SD)	EG Range
Pretest	12.35 (2.64)	5.00–19.00	12.22 (3.15)	5.00–20.00
Posttest	18.60 (1.90)	13.00–22.00	22.53 (3.03)	15.00–28.00
Learning gains	6.25	3.00–9.00	10.31	5.00–14.00

Note. CG=Control Group, EG=Experimental Group. $N=60$ per group

experimental groups, adhering to STPFSL procedures as detailed by (Brown & Singh, 2021; Brundage & Singh, 2023b). The primary objective was to assess ChatGPT's impact on conceptual understanding of thermodynamic principles and on reducing common misconceptions. To achieve this, we employed PCA, K-Means Clustering, and Heatmaps to analyze learning gains, changes in conceptual understanding, persistent misconceptions, and the intervention's effectiveness within the CILP framework.

Initial assessments and conceptual learning gains analysis

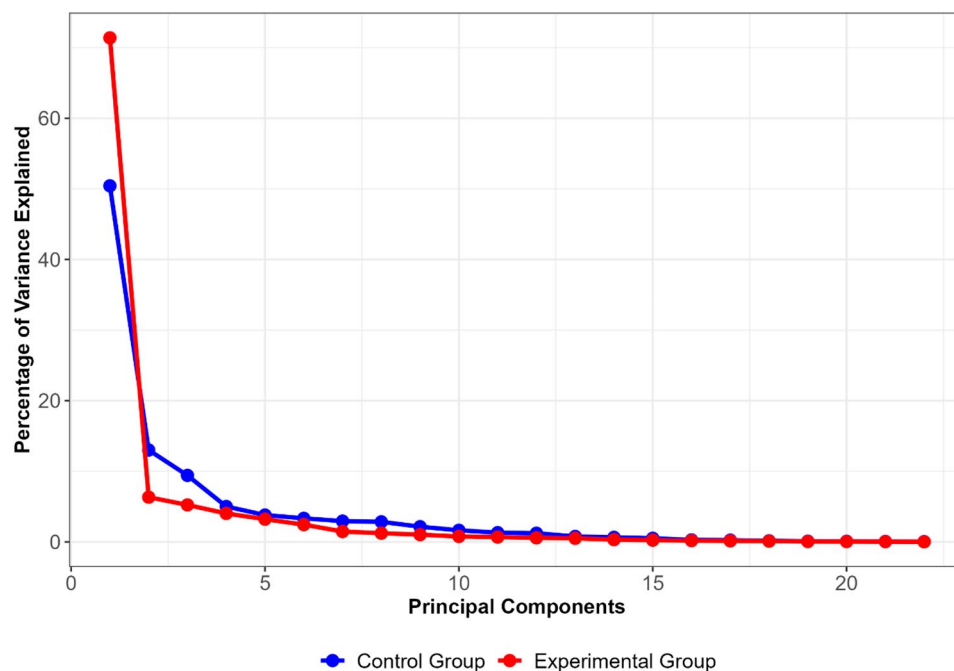
Pretest results revealed no statistically significant differences between the control and experimental groups (Table 2). However, post-intervention, the experimental group achieved higher posttest scores ($M=22.53$, $SD=3.03$) compared to the control group ($M=18.60$, $SD=1.90$). The experimental group also showed greater learning gains (10.31 vs. 6.25 for the control group),

indicating ChatGPT's potential to enhance conceptual learning in thermodynamics.

A Mann–Whitney U test confirmed a statistically significant difference in posttest scores between the control and experimental groups ($U=518.50$, $p<0.001$, Cohen's $d=0.614$). Additionally, a Wilcoxon signed-rank test demonstrated significant within-group improvements from pretest to posttest ($W=0$, $Z=-9.506$, $p<0.001$), with the experimental group showing notably larger gains.

Dimensional analysis of conceptual learning gains

To further explore factors contributing to learning gains, we performed PCA with Promax rotation. Parallel Analysis indicated the retention of three components in the control group, whereas it supported retaining only one component in the experimental group (see Fig. 1). The results of the Parallel Analysis, including comparisons of observed and simulated eigenvalues, are presented in Appendix B (Figures B5–B6). In the control group, three components explained 70.6% of the variance, reflecting conceptual themes of Ideal Gas Laws, Entropy, and Cyclic Processes. Conversely, the experimental group displayed a single dominant component labeled General Thermodynamics Understanding, accounting for 71.4% of the variance (Fig. 1). This single dominant component is pedagogically significant, as it suggests a more integrated conceptual structure potentially facilitated by ChatGPT's iterative feedback, aligning with the CILP framework to encourage broader conceptual linkages, such as

**Fig. 1** Explained variance by retained PCA components for control and experimental groups

connecting work, heat, and internal energy. This unified conceptual development contrasts with the fragmented topic-specific understanding observed in the control group.

The loadings matrix (Appendix B, Table B1) identifies thermodynamic topics most strongly associated with each factor, offering a basis for comparing conceptual integration across groups. These results suggest that ChatGPT, in combination with the CILP framework, played a role in fostering a cohesive understanding of thermodynamic concepts among students in the experimental group. The observed difference in factor structure supports the proposition that iterative feedback provided by ChatGPT promoted a more unified conceptual framework within the experimental group.

Patterns and trends analysis

Building on the findings from the dimensional analysis, we examined trends in students' overall conceptual understanding to explore group-level patterns. K-means clustering was applied to students' pretest thermodynamics scores, resulting in the identification of three optimal clusters: Low, Moderate, and High levels of understanding (see Fig. 2). The Elbow method and Silhouette Analysis confirmed the three-cluster solution as providing the best fit.

Heatmap analysis (Fig. 3) highlighted distinct patterns of improvement between groups. While the control group showed moderate post-intervention gains, with higher scores in Adiabatic Processes (80.80%) and Cyclic Processes (56.87%), their performance in topics like Engine Efficiency (36.40%) and Reversible Processes

(39.91%) remained relatively low. Conversely, the experimental group demonstrated significant improvements across a broader range of topics, particularly in Relation to PV and PV Diagrams, achieving scores above 90%.

The experimental group consistently outperformed the control group across most topics, as evident in the heatmaps. Significant gains were observed in the Low Understanding cluster of the experimental group, particularly in complex topics such as Internal Energy (52.21–66.27%) and Adiabatic Processes (56.87–72.35%). However, both groups showed limited improvement in Engine Efficiency and Relation to QT, suggesting the need for more targeted prompts to address persistent misconceptions.

Misconceptions analysis

Chi-square tests revealed significant reductions in misconceptions for both groups, with the experimental group showing greater overall improvement (Appendix B, Table B5). For example, the control group reduced misconceptions in Isobaric ($\chi^2(119)=150.71$, $p=0.026$) and Adiabatic Processes ($\chi^2(119)=149.18$, $p=0.032$). In contrast, the experimental group demonstrated broader reductions, particularly in Irreversible Processes ($\chi^2(119)=236.84$, $p<0.001$) and Entropy ($\chi^2(119)=186.49$, $p<0.001$). These results suggest that ChatGPT was more effective in addressing misconceptions in complex topics. Both groups showed limited improvement in Reversible Processes and Engine Efficiency, highlighting the need for more refined instructional strategies. Additional interventions, such as targeted, multi-step prompts and real-world applications,

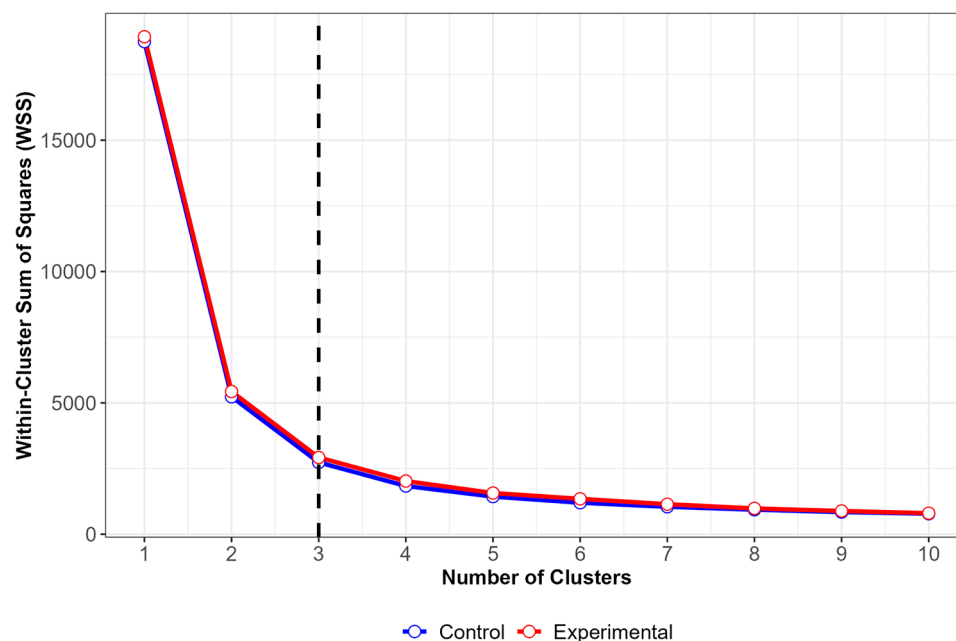


Fig. 2 Determination of optimal clusters in pretest control and experimental groups using the Elbow Curve method

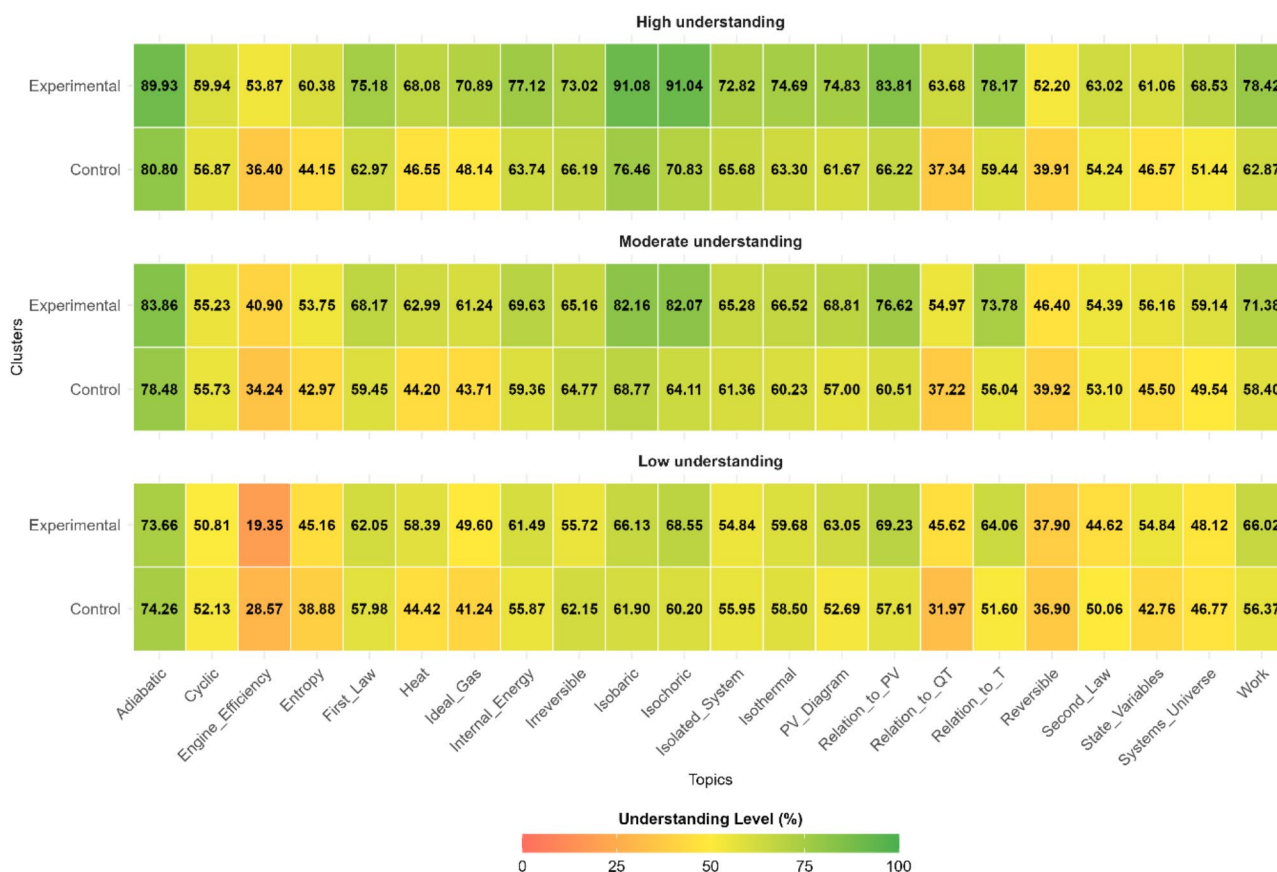


Fig. 3 Heatmap analysis of posttest conceptual understanding in the control and experimental groups

may be necessary to address the inherent complexity of these topics effectively.

The chi-square test results demonstrate that the ChatGPT intervention significantly reduced misconceptions in the experimental group, with p -values < 0.001 across most dimensions. This impact is attributed to ChatGPT's ability to provide immediate, tailored feedback, enabling students to refine their understanding of complex concepts such as Entropy, Internal Energy, and thermodynamic relationships. Although traditional methods also yielded improvements in the control group, they were less consistent and effective, particularly for topics requiring higher-order thinking and application.

Misconceptions segmentation

To investigate how misconceptions persisted or were resolved, a secondary K-means clustering analysis was conducted on students' misconception scores, resulting in three distinct clusters: High, Moderate, and Low Misconception levels within both control and experimental groups. The optimal number of clusters was determined using the Elbow method, and the validity of the cluster solution was confirmed by Silhouette Analysis (Appendix B, Figures B7-B8). This segmentation enabled an examination of how students with varying initial misconception

levels responded to the CILP-aligned intervention. Students with fewer misconceptions benefited from foundational prompts, whereas students exhibiting more extensive misconceptions required advanced prompt structures. Table B6 (Appendix B) provides complementary descriptive statistics for principal component scores across clusters and groups.

In the control group (Fig. 4), misconceptions persisted prominently on key topics such as the Ideal Gas Law and Entropy. The High Misconception Cluster ($n=40$) showed minimal improvement from pretest to posttest, particularly on Ideal Gas Law and Thermodynamic Processes, indicating traditional instruction inadequately addressed persistent misconceptions. The Moderate Cluster ($n=53$) exhibited slight progress, reflecting the limited effectiveness of conventional methods. The Low Cluster ($n=27$) demonstrated marginal gains, suggesting traditional instruction was more suitable for students with fewer initial misconceptions.

Figure 4 depicts two principal components: Ideal Gas and Thermodynamic Processes (x-axis) and Entropy and Systems (y-axis). Positions along the x-axis represent students' understanding of ideal gas laws and thermodynamic processes, while the y-axis corresponds to understanding entropy and system-level properties. Dashed

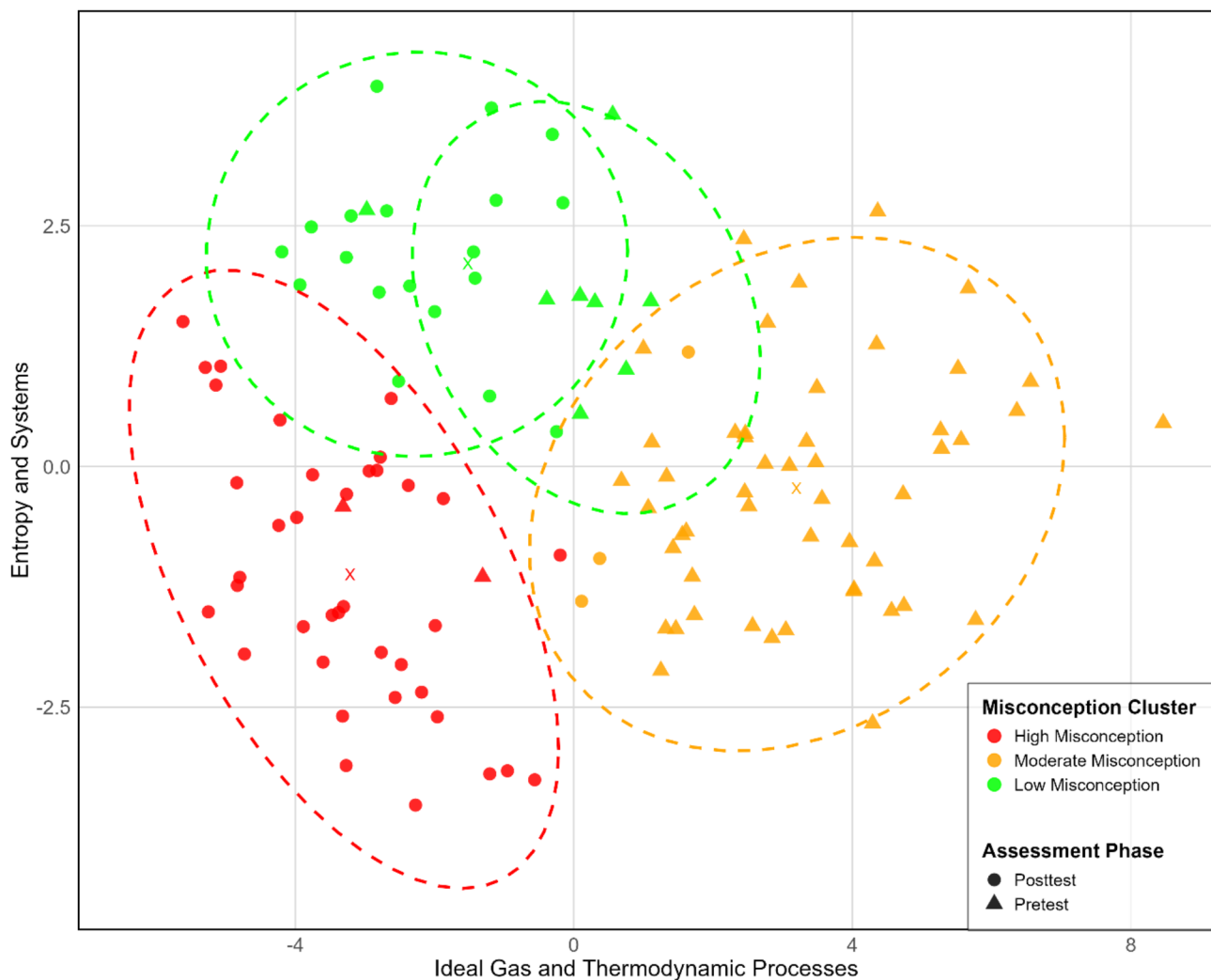


Fig. 4 Student misconceptions segmentation in the control group

ellipses represent misconception clusters with centroids (“X”) marking mean cluster positions. Triangles denote pretest scores, and circles indicate posttest results. The High Misconception Cluster (red ellipse) displayed minimal shifts, and the Moderate (orange ellipse) and Low (green ellipse) Clusters similarly showed limited progress.

In contrast, the experimental group (Fig. 5) demonstrated significant reductions in misconceptions following the ChatGPT-based interventions, particularly within the Moderate and Low Clusters. The High Misconception Cluster ($n=24$) also improved compared to the control group, although a subset required further instructional support. The Moderate Cluster ($n=45$) showed notable improvement in complex thermodynamic topics, while the Low Cluster ($n=51$) strongly benefited from iterative, feedback-rich prompts provided by ChatGPT.

Figure 5 plots two principal components: General Thermodynamics Understanding (x-axis), integrating

various thermodynamic topics into a unified conceptual dimension, and a Second Principal Component (y-axis), distinguishing misconceptions related to entropy and reversible processes. The Moderate (orange ellipse) and Low (green ellipse) Clusters shifted rightward from pretest (triangles) to posttest (circles), indicating robust conceptual gains. Although the High Misconception Cluster (red ellipse) remained the most resistant, its centroid moved positively along the x-axis, reflecting meaningful progress.

In conclusion, integrating ChatGPT with CILP-aligned strategies effectively reduced thermodynamics misconceptions, particularly for students initially classified within moderate or low misconception clusters. Iterative AI-driven feedback facilitated deeper conceptual integration compared to traditional methods. Persistent difficulties in complex topics (e.g., Adiabatic Processes, Entropy, Engine Efficiency) among students in the High Misconception cluster underscore the need for refined prompt

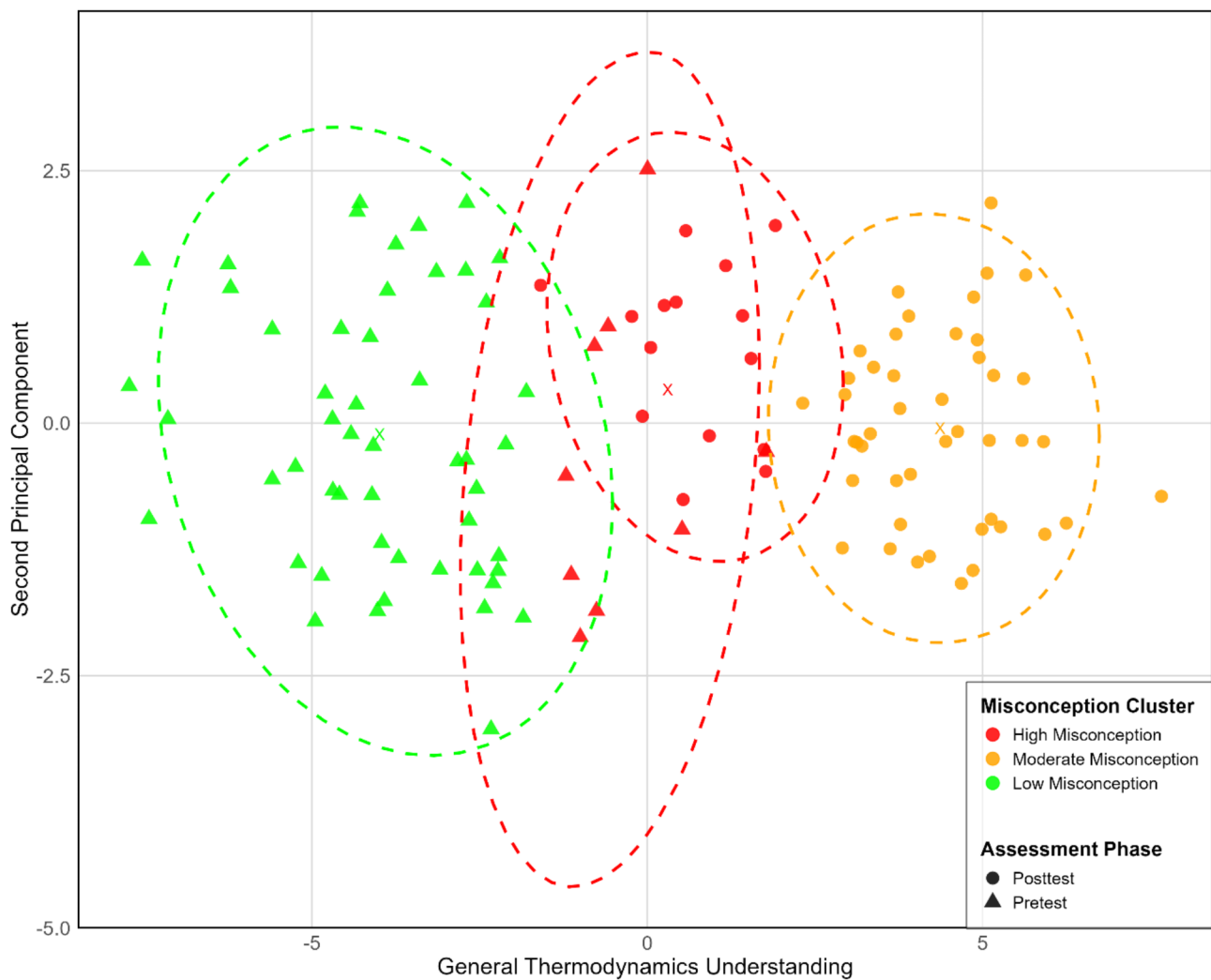


Fig. 5 Student misconceptions segmentation in the experimental group

engineering and targeted instructional scaffolds. Future implementations should further optimize these interventions, adapting the CILP framework to diverse learner needs to enhance instructional effectiveness through generative AI.

For RQ2

To address Research Question 2, this study investigated how Moroccan engineering students perceive the role of ChatGPT in enhancing their understanding of thermodynamics, focusing on usability, instructional support, and overall effectiveness. Data were collected through the TALES, providing insights into ChatGPT's classroom impact. Key findings are summarized in Table 3, which outlines the relationship between ChatGPT usage frequency and its perceived utility, highlighting significant trends in students' experiences and acceptance of this technology.

A series of Mann-Whitney U tests were conducted to compare the perceptions of High-Frequency (HF) and Low-Frequency (LF) ChatGPT users across several dimensions. A statistically significant difference was found in the helpfulness dimension, where LF users rated ChatGPT as more helpful ($M = 4.27$, $SD = 0.594$) compared to HF users ($M = 3.93$, $SD = 0.495$), $U = 240.5$, $p = 0.037$. No significant differences were found in other dimensions, such as ease of use, misconception addressing, or overall experience. This suggests that ChatGPT's utility remains stable, regardless of usage frequency.

These findings suggest that the quality of interaction is more important than frequency for maximizing ChatGPT's benefits. LF users likely experienced a novelty effect, enhancing their cognitive engagement with the tool, while HF users may have encountered cognitive overload or habituation, which reduced their perception of ChatGPT's helpfulness.

Table 3 Comparative analysis of perceptions on ChatGPT effectiveness by usage frequency

Variables	Category	Mean	SD	U	p-value
Ease of use	HF	4.16	0.737	336	0.985
	LF	4.20	0.561		
Helpfulness	HF	3.93	0.495	240.5	0.037*
	LF	4.27	0.594		
Misconception addressing	HF	3.93	0.618	353	0.755
	LF	3.87	0.352		
Satisfaction with explanations	HF	4.16	0.562	284	0.287
	LF	4.33	0.617		
Overall experience	HF	4.22	0.517	338	1.000
	LF	4.20	0.676		
Recommend to peers	HF	4.31	0.596	274.5	0.226
	LF	4.53	0.516		
Future use	HF	4.47	0.548	385	0.357
	LF	4.33	0.488		
Compared to other methods	HF	4.13	0.457	336.5	0.991
	LF	4.13	0.516		

Note. HF=High Frequency, LF=Low Frequency, M=Mean, SD=Standard Deviation, U=Mann-Whitney U Statistic, *p-value<0.05 indicates statistical significance

A visual representation (Fig. 6) illustrates student perceptions across several dimensions, ranging from negative to very positive.

The majority of students rated ChatGPT positively, with 31.67% rating it as Very Positive and 55% as Positive, highlighting its user-friendly interface and interactive capabilities, which enhanced the learning experience. Additionally, 71.67% of students praised its informative assistance, emphasizing its value in supporting learning. However, responses regarding misconception addressing were more varied, indicating opportunities for improvement, particularly in handling complex misconceptions that are harder to resolve through surface-level feedback.

Furthermore, 41.67% of students expressed a strong intent to recommend ChatGPT, and its anticipated future use highlights its potential as a valuable educational tool. While ChatGPT's strengths in usability and content delivery were evident, linking student intent to recommendation with learning outcomes would provide further insights into its overall effectiveness.

For RQ3

Research Question 3 examined the broader impacts of integrating ChatGPT into classroom settings, specifically investigating how different modes of student engagement with the AI chatbot influenced their understanding of thermodynamics and ability to rectify misconceptions. The analysis involved two complementary approaches: (1) correlation analyses (Spearman's rho) to identify patterns among ChatGPT usage, subjective learning

perceptions, and learning gains, and (2) a GLM regression to further quantify these relationships while controlling for initial understanding (Table 4).

The integrated analyses revealed consistent patterns. Both correlation and regression results demonstrated a significant negative relationship between initial understanding and learning gains ($\rho = -0.402$, $p < 0.01$; $B = -0.24$, $p = 0.008$), indicating that students with lower initial understanding benefited most substantially from ChatGPT-enhanced instruction within the CILP framework. This finding reinforces the framework's effectiveness in addressing knowledge gaps and supporting students with limited prior knowledge.

Regarding interaction quality, both analyses identified session duration as positively correlated with improved learning outcomes. Although the correlation analysis indicated a positive but non-significant trend ($\rho = 0.182$, $p = 0.087$), the regression analysis revealed a statistically significant relationship ($B = 0.55$, $p = 0.038$) after controlling for other variables. Thus, extended and structured interactions with ChatGPT under the CILP framework significantly enhanced conceptual learning gains.

Conversely, neither correlation nor regression analyses found significant relationships between ChatGPT usage frequency, perceived helpfulness, or perceived effectiveness in addressing misconceptions and students' learning gains (all p s > 0.05). This finding underscores the importance of interaction quality, characterized by sustained and structured inquiry, rather than interaction frequency alone.

Finally, the analysis of the control group showed that initial understanding was not significantly associated with learning gains ($B = -0.11$, $p = 0.339$). This confirms that the learning advantage for students with lower initial understanding was unique to the experimental group. Overall, these results highlight ChatGPT's potential as an educational tool, particularly when integrated into inquiry-based and constructivist learning environments within the CILP framework.

Discussion and implications

This study examines the integration of ChatGPT, a generative AI chatbot, as a supplementary learning tool in an introductory thermodynamics course for first-year Moroccan engineering students. The primary objectives were to enhance conceptual understanding, address common misconceptions, and foster student engagement within a structured, interactive learning environment. The findings underscore the potential of generative AI tools to enhance STEM education through real-time, personalized feedback and active student participation. This study contributes to the field by demonstrating the application of ChatGPT in Moroccan engineering classrooms, highlighting its potential to address regional

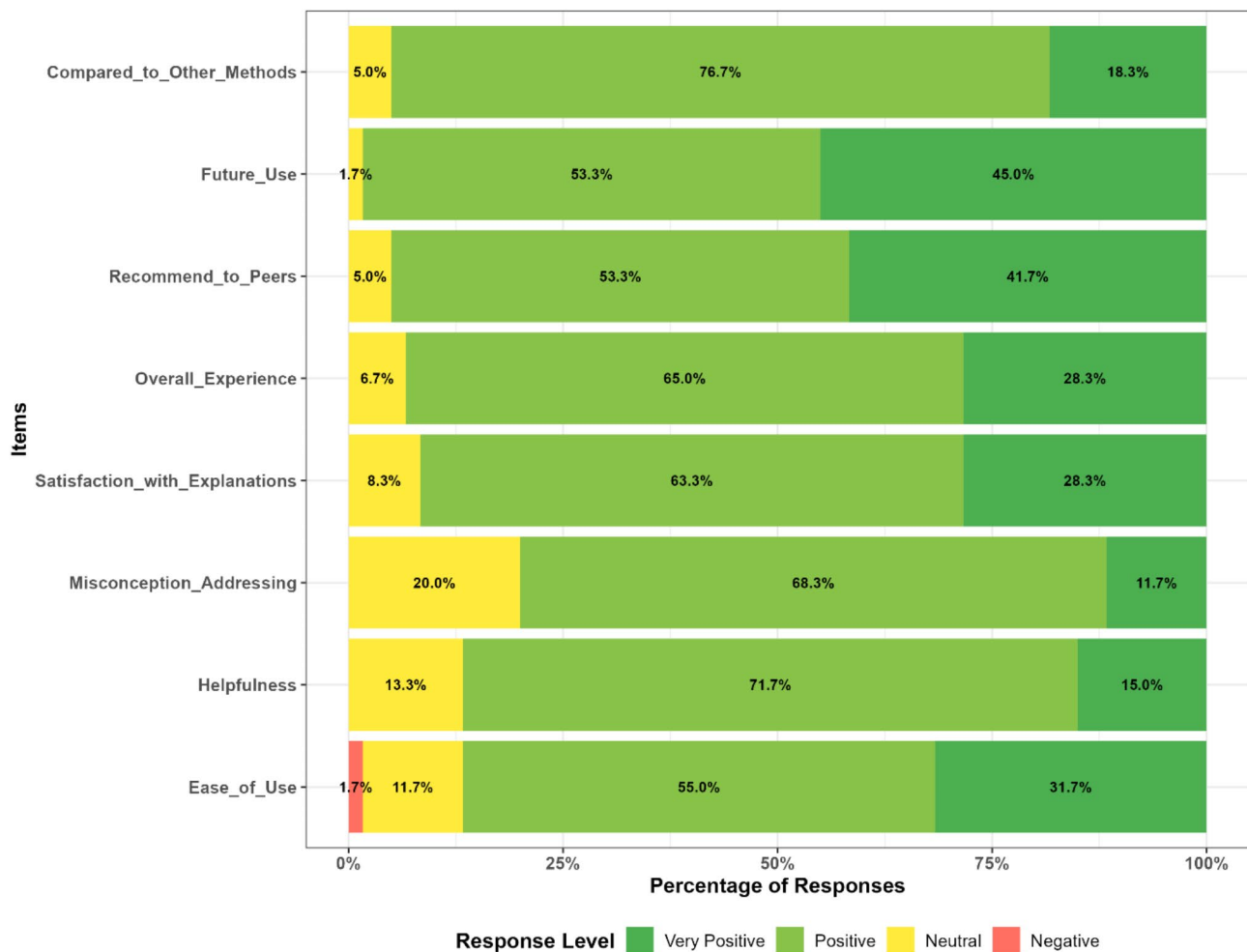


Fig. 6 Comparative perceptions of ChatGPT pedagogical effectiveness

Table 4 Predictors of student understanding and misconception correction in thermodynamics topics

Predictor	B	SE	t	p-value
Intercept	19.15	4.53	4.23	<0.001***
Frequency of use	-0.16	0.28	-0.58	0.562
Session duration	0.55	0.26	2.13	0.038*
Ease of use	-0.17	0.40	-0.44	0.664
Helpfulness	-0.36	0.51	-0.71	0.483
Misconception addressing	0.04	0.46	0.09	0.927
Confidence improvement	-0.46	0.49	-0.94	0.354
Trust in accuracy	-0.49	0.49	-1.01	0.320
Initial understanding (EG)	-0.24	0.09	-2.73	0.008**
Initial understanding (CG)	-0.11	0.11	-0.97	0.338

Note. CG=Control Group, EG=Experimental Group, B=Unstandardized regression coefficient, SE=Standard Error, t=t-value. Significance levels of p-values: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

educational challenges and its role in advancing STEM education globally.

Enhancing conceptual Understanding

The findings demonstrate that the integration of ChatGPT significantly enhances students' conceptual understanding of thermodynamics, particularly in challenging topics such as adiabatic and isothermal processes. Students in the experimental group achieved notably greater learning gains (10.31 compared to 6.25 in the control group) and developed a more cohesive understanding of thermodynamic principles. The PCA revealed that the experimental group's conceptual understanding was characterized by a single dominant component, "General Thermodynamics Understanding," accounting for 71.4% of the variance. This highlights ChatGPT's ability to foster integrated understanding by connecting key concepts, such as work, heat, and internal energy, in contrast to the fragmented understanding observed in the control group.

These findings align with prior studies emphasizing the effectiveness of personalized AI feedback in supporting the development of mental models (Brundage & Singh, 2023a; Wheeler & Scherr, 2023). The CILP Framework guided students to iteratively refine their prompts,

analyze ChatGPT's feedback, and reflect on their learning. This inquiry-driven approach facilitated deeper conceptual connections and aligned with constructivist teaching theories that prioritize meaningful, student-centered engagement (Sreelohor et al., 2022).

ChatGPT's natural language interaction and step-by-step explanations enabled students to engage with the material at their own pace. For instance, when students encountered difficulties with concepts such as entropy or internal energy, ChatGPT provided detailed, customized explanations, offering real-time clarifications that are often impractical in traditional classroom settings (Chen et al., 2023). Furthermore, this study extends previous research by demonstrating that generative AI not only improves conceptual understanding but also enhances self-regulated learning through iterative feedback. This finding illustrates the potential of AI tools to accommodate diverse learning styles and proficiency levels, particularly in complex subjects like thermodynamics (Kieser et al., 2023; Kieser & Wulff, 2024).

Despite these benefits, limitations persist. Challenges in quantitative topics, such as engine efficiency and reversible processes, highlight the need for more advanced AI capabilities to support complex problem-solving tasks. These challenges align with previous research highlighting AI limitations in numerical reasoning (Demirezen et al., 2023; Aldazharova et al., 2024). Future research should explore how generative AI can better support advanced quantitative problem-solving, particularly in engineering contexts.

Despite ChatGPT's strengths in providing clear conceptual explanations, its limitations remain evident in quantitative problem-solving, particularly in topics such as engine efficiency and reversible processes. Recent advancements in LLMs, notably DeepSeek-R1, a newly developed, open-source generative AI system specializing in structured, step-by-step reasoning, have demonstrated promising capabilities in advanced mathematics, symbolic computation, and logical problem-solving (DeepSeek-AI et al., 2025; Tewari & Tewari, 2025). By externalizing assumptions, intermediate formulas, and numerical results, DeepSeek-R1 offers an accessible alternative to support students' quantitative reasoning and reduce cognitive load during complex calculations (DeepSeek-AI et al., 2025). Although DeepSeek-R1 may overcome some of ChatGPT's quantitative limitations, further empirical validation is needed to assess its effectiveness in educational settings.

To address these limitations, hybrid instruction models, in which AI tools are supplemented by teacher-led discussions, may help identify and correct misunderstandings early. Leveraging AI tools like ChatGPT requires significant investments in curriculum redesign and educator training to ensure effective implementation

in STEM education (Bernabei et al., 2023; Park et al., 2023). Aligning AI tools with traditional educational goals necessitates targeted professional development to prepare educators for AI-enhanced teaching environments (Dare et al., 2021; Gregorcic & Pendrill, 2023). For broader global impact, further exploration of how these strategies can be adapted for diverse educational systems and resource-limited settings is crucial. A hybrid instructional approach, combining ChatGPT's personalized feedback with teacher-led discussions, could further enhance learning outcomes (Perkowitz, 2023).

Addressing and reducing misconceptions

This study demonstrated that ChatGPT effectively reduces misconceptions in qualitative areas, such as entropy, internal energy, and the second law of thermodynamics. Students in the experimental group exhibited significantly greater reductions in misconceptions (e.g., regarding irreversible processes, $p < 0.001$) compared to the control group, with notable improvements in abstract topics like PV diagrams and pressure-volume relationships. These findings align with prior research emphasizing the effectiveness of generative AI in addressing conceptual misunderstandings and fostering integrated conceptual frameworks in STEM education (Banda & Nzabahimana, 2023; Kieser & Wulff, 2024; Modran et al., 2024).

The CILP Framework played a critical role in addressing misconceptions by encouraging students to iteratively refine their prompts and critically evaluate ChatGPT's feedback. By aligning this process with course-specific learning objectives, the framework supported self-regulated learning and reflection, enabling students to identify and correct misconceptions. For example, a student who initially misunderstood PV diagrams as representing linear relationships between pressure and volume was guided by ChatGPT's iterative explanations and visual analogies, eventually developing an accurate conceptual understanding. This aligns with previous studies emphasizing the importance of metacognitive skills in overcoming misconceptions (Kulgemeyer & Wittwer, 2023; Machost et al., 2024).

Heatmap analysis revealed that the experimental group exhibited significantly greater improvements across a broader range of topics, particularly in areas such as PV diagrams and pressure-volume (PV) relationships. To contextualize these findings, K-means cluster analysis identified three optimal subgroups of students based on their initial knowledge states or misconception patterns. This approach allowed us to track how these clusters of learners responded to the instructional intervention. For example, students in high-misconception clusters benefited from tailored scaffolding prompts that simplified foundational thermodynamics concepts, while those

in low-misconception clusters improved through more advanced and exploratory prompts. However, persistent misconceptions in areas such as engine efficiency and reversible processes suggest that achieving deeper conceptual change requires more than feedback alone, particularly for advanced learners.

These results underscore the importance of clustering in identifying the specific needs of diverse student subgroups. They also highlight how generative AI tools like ChatGPT can be adapted to target misconceptions across varying levels of conceptual readiness. Integrating AI tools into a broader pedagogical framework is critical, particularly for addressing complex topics where stand-alone AI interventions may fall short. To maximize the effectiveness of AI tools like ChatGPT, human oversight and instructional guidance are essential to prevent the reinforcement of errors (Kulgemeyer & Wittwer, 2023; Camacho-Lie et al., 2024; Machost et al., 2024). Furthermore, developing AI-supported curricula and providing teacher training programs are crucial for bridging the gap between AI capabilities and classroom objectives (Shernoff et al., 2017; Gregorcic et al., 2024).

Impact on student engagement and learning experience

This study demonstrated that ChatGPT positively influences student engagement and learning experiences, particularly when used strategically. Data from the TALES revealed that 71.67% of students praised ChatGPT for its ease of use and informative assistance, highlighting its role in fostering positive learning experiences. Interestingly, LF users reported higher satisfaction ($M = 4.27$) than HF users ($M = 3.93$, $p = 0.037$), suggesting that ChatGPT is most effective when used in moderation to avoid potential cognitive overload.

Excessive reliance on tools like ChatGPT may contribute to cognitive overload, diminishing their effectiveness over time (Lo et al., 2024). This finding aligns with cognitive load theory, which posits that learners benefit most when instructional materials are introduced at an optimal pace. Excessive or unmoderated use of generative AI tools, as noted by Ding et al. (2023), can lead to disengagement, further reinforcing the critical role of educator oversight.

The strategic integration of ChatGPT enhanced student engagement while fostering critical thinking and higher-order learning. Students who used ChatGPT for focused problem-solving tasks reported higher satisfaction, underscoring the importance of balancing AI-enhanced learning with traditional teaching methods. Incorporating student feedback into the design and implementation of AI tools is crucial for fostering positive learning experiences. Factors such as perceived ease of use, usefulness, and overall helpfulness strongly influence students'

willingness to engage with ChatGPT (Bernabei et al., 2023; Strzelecki, 2023).

This study underscores the transformative potential of ChatGPT as a supplementary tool for teaching thermodynamics, addressing both conceptual understanding and students' acceptance of AI. However, its adoption presents challenges related to trust, ethical implications, and limitations in quantitative reasoning (Costa et al., 2024; Elmas et al., 2024). The variability in ChatGPT's effectiveness, influenced by factors such as students' initial knowledge levels, suggests that AI integration must be context-sensitive and tailored to meet the specific needs of diverse learners. Rather than indiscriminately increasing AI tool usage, targeted and context-driven use of AI tools is critical for maximizing their educational benefits (Relmasira et al., 2023; Elmas et al., 2024).

Gamification approaches and simulation-based tools, such as PhET simulations, have proven effective in boosting student motivation and facilitating interactive visual explorations of phenomena (Liu & Israel, 2022; Diab et al., 2024). However, these methods often struggle to address intricate misconceptions in real-time without additional human scaffolding. In contrast, ChatGPT provides immediate, personalized, dialogue-based feedback that identifies and corrects students' mental model gaps, dynamically adjusting instructional support as misconceptions arise (Elmas et al., 2024; Kotsis, 2024). Integrating ChatGPT with established gamification and simulation methods could bridge the gap between student engagement and deeper conceptual understanding, two critical dimensions that are often not equally prioritized in purely gamified or simulation-driven interventions.

Effective strategies involve leveraging ChatGPT for pre-class assignments, formative assessments, and in-class problem-solving sessions to complement traditional teaching methods. Collaboration among educators, policymakers, and researchers is essential for developing best practices for AI integration that meet diverse learning needs (Machost et al., 2024). Future research should explore how generative AI can be optimized for different educational contexts and how it can complement traditional teaching methods to enhance student learning outcomes.

Conclusions

This study highlights the transformative potential of generative AI tools, especially ChatGPT, as supplementary resources in STEM education, particularly in introductory thermodynamics. By integrating ChatGPT with traditional teaching methods and leveraging the structured CILP Framework, students demonstrated significant learning gains, particularly in complex qualitative topics such as entropy and pressure-volume relationships.

ChatGPT's ability to provide personalized, real-time feedback played a critical role in addressing misconceptions and fostering deeper conceptual understanding, bridging gaps that traditional methods often struggle to fill.

The findings are particularly relevant in resource-constrained contexts like Morocco, where challenges such as high student-to-teacher ratios and limited access to advanced educational technologies are prevalent. ChatGPT offers scalable and adaptive learning opportunities that enhance student engagement and motivation in demanding STEM disciplines. Positive feedback from students reinforces its value as a supplementary pedagogical tool, highlighting its ease of use and effectiveness in supporting conceptual learning. However, ChatGPT's limitations in addressing quantitative reasoning and advanced problem-solving underscore the need for hybrid instructional approaches that combine AI tools with teacher-led interventions and complementary technologies, such as PhET simulations or virtual reality labs.

To maximize its effectiveness, instructors should adopt structured strategies such as the CILP Framework, which encourages iterative prompt refinement, critical evaluation of AI responses, and reflective learning. Collaborative activities that blend AI interaction with peer learning can further enhance student engagement and understanding. Additionally, professional development programs for teachers are crucial to ensure the ethical and strategic integration of generative AI tools into classrooms. These programs can help educators design effective AI-supported learning activities while addressing potential challenges, such as reinforcing misconceptions or fostering over-reliance on AI.

In conclusion, ChatGPT holds significant promise for enhancing conceptual understanding, reducing misconceptions, and fostering engagement in STEM education. Its effectiveness, however, relies on thoughtful implementation, continuous oversight, and the integration of complementary tools and strategies to address its current limitations. By adopting these approaches, educators can harness generative AI tools as a powerful complement to traditional teaching methods, advancing accessible and engaging STEM education globally.

Study limitations

This study has several limitations. The quasi-experimental design and convenience sampling from a single Moroccan institution limit the generalizability of the findings. The specific context of Moroccan engineering students, with unique curricula, language (French and Arabic), and cultural factors, may affect the applicability of results to other settings. Participants' familiarity with technology and generally positive attitudes toward AI may not reflect the broader student population, which

introduces potential bias. The short duration of one semester restricts the assessment of long-term learning outcomes. Furthermore, reliance on the free version of ChatGPT may have limited student 'deeper engagement with complex topics like thermodynamics. Additionally, the study's setting in a technology-enhanced classroom with reliable internet access may not represent other educational environments in Morocco, affecting the feasibility of implementing similar AI-based interventions elsewhere. Thus, caution is advised when generalizing the results, and further research is needed across diverse contexts.

Future research directions

Future research should focus on advancing generative AI algorithms to enhance quantitative reasoning, numerical problem-solving, and advanced calculations, further expanding their role in STEM education. Comparative studies of generative AI tools (e.g., ChatGPT vs. DeepSeek-R1) across diverse student populations and STEM curricula could clarify their relative strengths, contextual effectiveness, and potential for integration into evidence-based teaching and learning frameworks. Investigating hybrid pedagogical models that integrate AI with complementary technologies, such as PhET simulations, virtual reality labs, and blended learning frameworks, could provide a more comprehensive learning experience. Longitudinal studies are needed to assess the long-term impact of AI integration on knowledge retention, skill development, and higher-order thinking, particularly in resource-limited educational environments. Such research could also examine how sustained AI use influences students' motivation, metacognitive development, and refinement of mental models over time.

Further exploration is required to determine whether the benefits of generative AI extend across other STEM disciplines, such as chemistry, biology, and mathematics. Additionally, adapting AI tools like ChatGPT to diverse learner needs, through customized prompts tailored to varying proficiency levels, cultural contexts, and learning styles, will be essential for ensuring equitable access, particularly in under-resourced settings. Finally, research should evaluate how professional development programs prepare educators to integrate generative AI effectively. By fostering ethical and strategic AI use, educators can better leverage these technologies to enhance instruction. Addressing these research gaps will not only optimize AI's role in STEM education but also promote more inclusive, adaptive, and engaging learning environments worldwide.

Abbreviations

AI	Artificial intelligence
CILP	Constructivist inquiry-based learning prompting

CPGE	Preparatory classes for higher education institutions (classes préparatoires aux grandes Écoles)
CTT	Classical test theory
GLM	Generalized linear model
HF	High-frequency
LF	Low-frequency
LLM	Large language model
MPSI	Mathematics, physics, and engineering sciences (mathématiques, physique et sciences de l'Ingénieur)
PCA	Principal component analysis
PCSI	Physics, chemistry, and engineering sciences (physique, chimie et sciences de l'Ingénieur)
PNIA	National programme for student learning assessment (programme national d'Évaluation des Acquis)
STPFSL	Survey of thermodynamic processes and first and second laws
STPFSL-Short	Short version of the survey of thermodynamic processes and first and second laws
TALES	Technology acceptance and learning enhancement survey
TAM	Technology acceptance model
UTAUT	Unified theory of acceptance and use of technology

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s43031-025-00125-z>.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

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Author contributions

TEF conceptualized the study, developed the methodology, led the investigation, administered the survey, organized data, conducted data analysis, and oversaw the writing and revision of the original manuscript while managing the overall project. SA was responsible for survey administration and data organization. HL revised the second research instrument. DL coordinated project planning and facilitated collaboration among team members. EMAL revised the primary research instrument.

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Not applicable.

Data availability

The datasets used or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Research ethics

This study was conducted in accordance with the ethical standards of the pedagogical administration of the National School of Applied Sciences at Ibn Tofail University. Informed consent was obtained from all participants, who were informed of their right to withdraw from the study at any time without consequences. All data were anonymized and securely stored in compliance with local data protection regulations to ensure confidentiality. The integration of ChatGPT was carefully monitored to ensure that no participant was disadvantaged; all students were provided equal access to the necessary technology. Participation in this study was voluntary and did not affect academic standing or grades.

Conflicts of interest

The authors declare no conflicts of interest.

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